Week 40: Loop colors

Adult guide built from the problem files · Unpiloted. The guide's cord pretest is unperformed.

This week's adult guide is week-40-facilitator.pdf, whose source is in editable/facilitator-src/. These problem files hold each problem's one-sentence idea, not its answers.

Problems and routes

Problem	K–1	2–3	4–5	route
How many whole arcs?	P1	P1	P1	first
Complete one crossing	P2			first
Every coloring of the three-crossing loop	P3	P2	P2	first
Colorings of the plain loop	P4	P3	P3	first
Which cords become the plain loop?	P5	P4	P4	first
Colorings before and after a two-crossing move		P5	P5	first
One completion for each choice		P6		return
Curls, slides and the count as a certificate			P6	return
Equal counts prove nothing			P7	return
Six end-color cases across two crossings	P6			return

First route. K–1 P1–P5; Grades 2–3 P1–P5; Grades 4–5 P1–P5. **Return visits.** K–1 P6; Grades 2–3 P6; Grades 4–5 P6–P7.

Key, one entry per problem

How many whole arcs?

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route

Idea. An arc runs from one underpass to the next, straight through its overpasses; the figure-eight has four.

Complete one crossing

K–1 P2 · first route

Idea. Two colors at a crossing force the third: the same color if they agree, the remaining color if not.

Every coloring of the three-crossing loop

K–1 P3 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. The trefoil has nine colorings: three one-color and the six orders of R, B, G.

Colorings of the plain loop

K–1 P4 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. A plain loop is one arc, so it has only the three one-color colorings.

Which cords become the plain loop?

K–1 P5 · Grades 2–3 P4 · Grades 4–5 P4 · first route · wording differs by band

Idea. Three cords open into a plain loop; the trefoil will not, and failing is not yet a proof.

Colorings before and after a two-crossing move

Grades 2–3 P5 · Grades 4–5 P5 · first route · wording differs by band

Idea. Each choice of end colors has exactly one coloring before the move and one after, with the same end colors.

One completion for each choice

Grades 2–3 P6 · return route

Idea. The crossing rule forces each color in turn, so a choice at L and P has only one completion.

Curls, slides and the count as a certificate

Grades 4–5 P6 · return route

Idea. A curl and a slide also match colorings one to one, so every move keeps the count, and 9 against 3 proves the three-crossing loop is not a plain loop.

Equal counts prove nothing

Grades 4–5 P7 · return route

Idea. The figure-eight has only the three one-color colorings but is knotted: equal counts do not show two loops are the same.

Six end-color cases across two crossings

K–1 P6 · return route

Idea. Fixed end colors work exactly when the bent strand’s ends agree and the straight strand’s ends agree.